

INTERIM REPORT OF

Transcriptome Sequencing

On Illumina Novaseq X Plus Platforms

Project ID: 260086

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1. PROJECT DETAILS & SAMPLE DETAILS

This interim report outlines the status of RNA extraction, quality control, and library preparation for the project 260086.

TABLE 1.1: PROJECT SPECIFICATIONS

PARAMETER	VALUE
Service Type	Transcriptome Sequencing
Platform	Illumina Novaseq X Plus
Read Length	2 X 150 bp PE
Data Throughput	~08-09GB / Sample
Type of Sample	Tissue
No. of Samples	11
Samples ID of Libraries Prepared	Control Brain, Control Heart, Control Muscle, Test Brain, Test Heart, Test Muscle
Shipping Condition	Dry Ice
No. of Libraries Prepared	11

2. METHODS

2.1 Isolation and Quantitative analysis of RNA:

RNA sample was extracted from tissue sample using QIAGEN RNeasy mini kit (CAT.NO:74106). RNA quantity was measured using Qubit® 4.0 fluorometer and quality were analyzed by using 1% agarose gel.

2.2 Preparation of library:

The paired-end sequencing library was prepared using NEBNext® Ultra™ RNA Library Prep Kit for Illumina. (NEB #E7770). The library preparation process was initiated with 100 ng input. Ribosomal RNA was removed using depletion was carried out using ribodepletion kit KAPA RNA HyperPrep Kit with RiboErase (HMR), Cat no: KK8561) as per user manual. Ribo-depleted RNA was subjected to fragmentation, first & second-strand cDNA synthesis, end-repair, 3' adapter ligation, selective enrichment of adapter-ligated DNA fragments through PCR amplification, followed by validation of Library on Agilent 4150 tape station. The final library was pooled with other samples, denatured & loaded on to flow cell. On the flowcell, cluster generation & sequencing was performed using Illumina Novaseq X plus platform to generate 2×150bp paired-end (PE) reads.

2.3 Quantity and quality check (QC) of library on Agilent Tape Station 4150:

The amplified library analyzed on TapeStation 4150 (Agilent Technologies) using High Sensitivity D1000 ScreenTape® as per manufacturer's instructions.

2.4 Cluster Generation and Sequencing:

After obtaining the Qubit concentration for the library and the mean peak size from Tape Station profile, library will be loaded onto Illumina Novaseq X Plus for cluster generation and sequencing. Paired-End sequencing allows the template fragments to be sequenced in both the forward and reverse directions. The library molecules will bind to complementary adapter oligos on paired-end flow cell. The adapters are designed to allow selective cleavage of the forward strands after re-synthesis of the reverse strand during sequencing. The copied reverse strand is then used to sequence from the opposite end of the fragment.

3. RESULTS

3.1 RNA QC using Agarose Gel

RNA quality was validated using 1% Agarose Gel electrophoresis to ensure intact ribosomal RNA bands (28S and 18S).

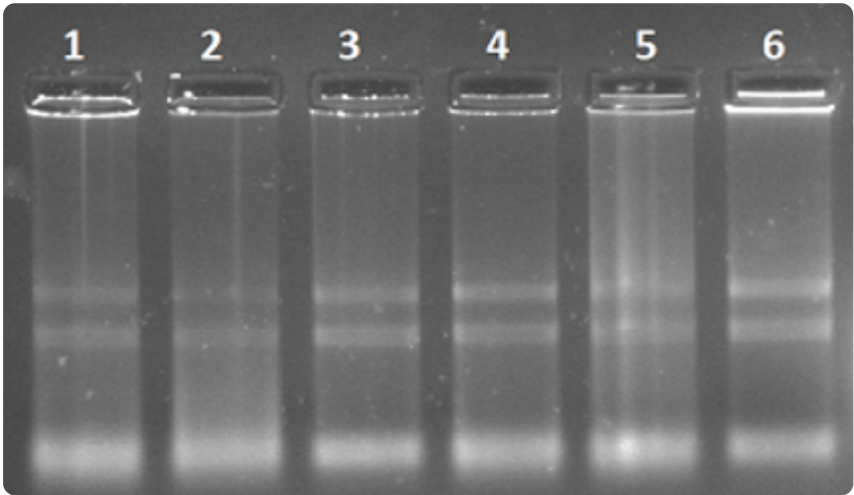


Image 01: RNA QC on 1% agarose gel

LANE ID TO SAMPLE NAME MAPPING

LANE ID	SAMPLE NAME	LANE ID	SAMPLE NAME	LANE ID	SAMPLE NAME
1	Control Brain	2	Control Heart	3	Control Muscle
4	Test Brain	5	Test Heart	6	Test Muscle

3.2 Qubit Quantification Data

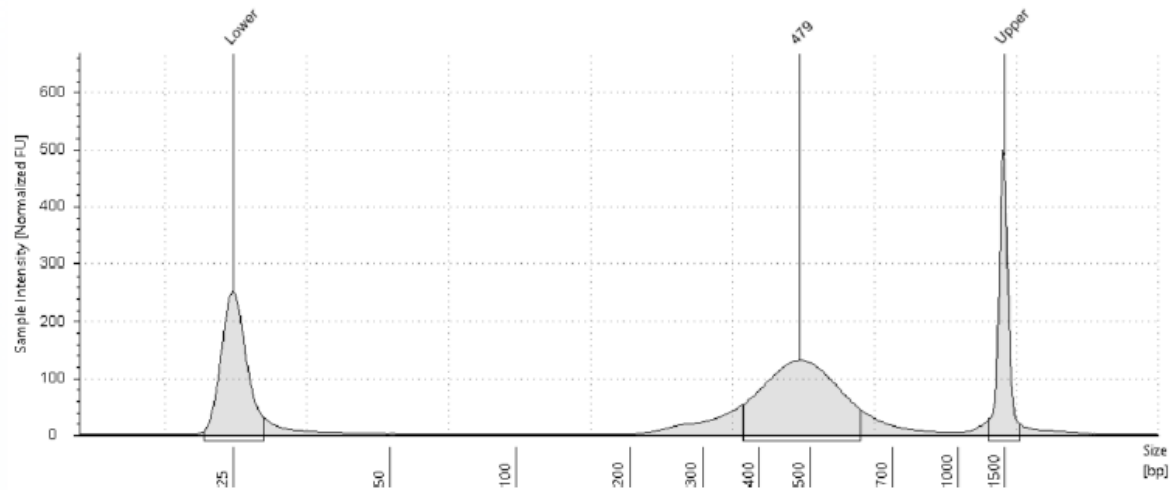
TABLE 3.2: RNA QUANTIFICATION (QUBIT)

S.N	SAMPLE ID	CONC. (NG/ML)	VOLUME (ML)	YIELD (MG)	REMARKS
1	Control Brain	43	50	2.15	QC PASS
2	Control Heart	20.2	50	1.01	QC PASS
3	Control Muscle	25.2	50	1.26	QC PASS
4	Test Brain	69.6	50	3.48	QC PASS
5	Test Heart	114	50	5.7	QC PASS
6	Test Muscle	45	50	2.25	QC PASS

4. QUANTITY AND QUALITY CHECK (QC) OF LIBRARY ON AGILENT TAPE STATION 4150

4.1 TapeStation 4150 Profiles of Library:

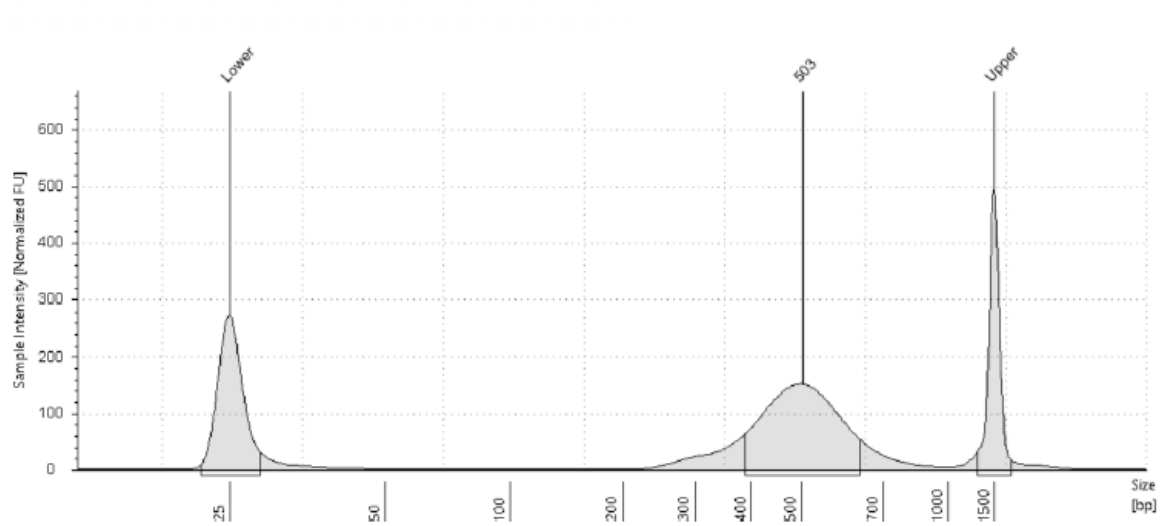
Agilent High Sensitivity D1000 ScreenTape® profiles for prepared libraries are shown below:



Sample Table

Well	Conc. [pg/ul]	Sample Description	Alert	Observations
C2	510			

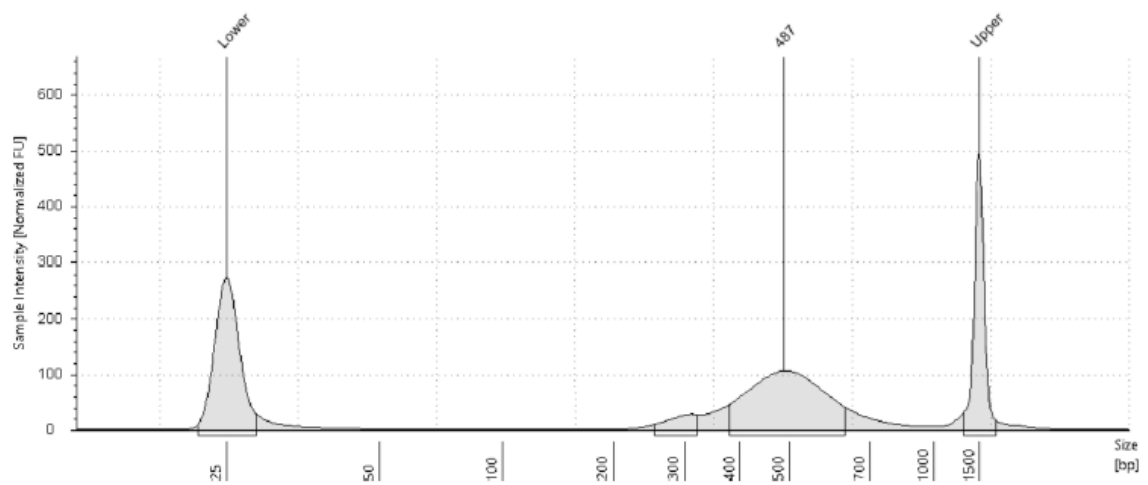
Profile 1: Control Brain



Sample Table

Well	Conc. [pg/ul]	Sample Description	Alert	Observations
D2	521			

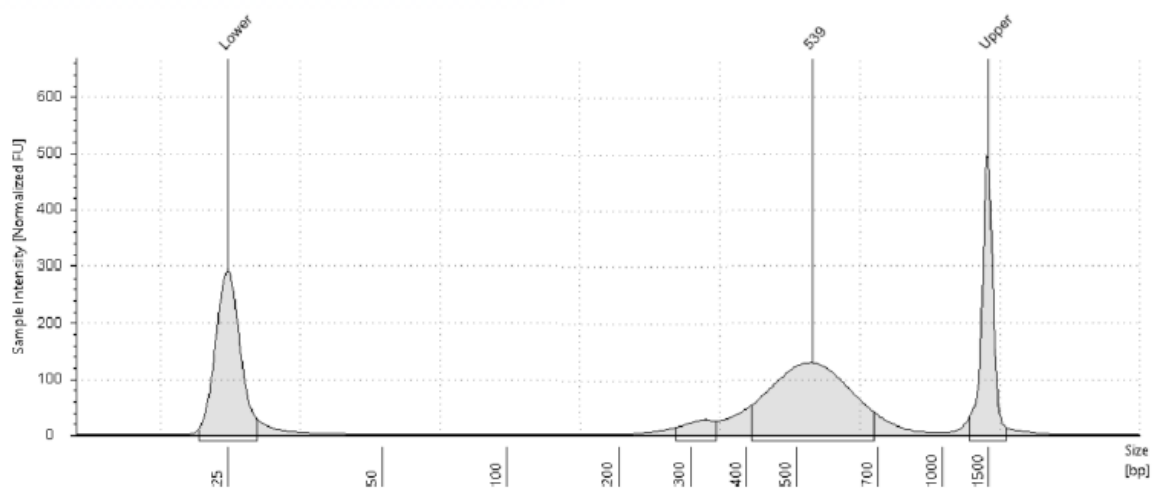
Profile 2: Control Heart



Sample Table

Well	Conc. [pg/ul]	Sample Description	Alert	Observations
E2	436			

Profile 3: Control Muscle



Sample Table

Well	Conc. [pg/ul]	Sample Description	Alert	Observations
F2	480			

Profile 4: Test Brain

5. CONCLUSIONS:

- The library was prepared from the samples using KAPA RNA HyperPrep Kit with RiboErase (HMR), Cat no: KK8561. - The average size of library is 342bp. - The library will be sequenced on Illumina Novaseq X plus platform using 2x150 bp PE chemistry.

[END OF INTERIM REPORT]